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UNIVERSITY
NAVI MUMBAI

Semester: III | **B. Tech. Artificial Intelligence and Data Science** (~~Common to all Branches Except CSBS~~) | **39524304**

2413ADU3T1 | **Embedded Systems**

Date: 01/12/2025 | **Time: 10:00 AM TO 12:00 NOON** | **Max Marks: 60**

Q.1	**Attempt all Questions**	**20 Marks**	**CO**	**BT**
a)	Perform the conversion of an octal number $(251)_8$ into decimal number.	4	CO1	BT4
b)	Compare Microprocessor and Microcontroller.	4	CO2	BT4
c)	Explain I/O port programming with suitable program examples.	4	CO4	BT3
d)	Investigate the case study of Washing Machine in Embedded Systems.	4	CO5	BT5
e)	Describe three stage pipelining in ARM microcontroller.	4	CO6	BT2

Q.2	**Solve any Four**	**20 Marks**	**CO**	**BT**
a)	Describe the working of De-Multiplexer with diagram and circuit.	5	CO1	BT2
b)	Discuss what do you mean by RISC and CISC architectures.	5	CO2	BT2
c)	Explain SJMP, AJMP, LJMP instructions in 8051 microcontroller.	5	CO3	BT2
d)	Write an assembly language to Multiply two 8-bit numbers.	5	CO4	BT4
e)	Draw the diagram for DC motor interfacing with 8051 and explain it with simple program.	5	CO5	BT4
f)	Draw and explain the main blocks of ARM architecture.	5	CO6	BT2

Q.3	**Solve any Two**	**20 Marks**	**CO**	**BT**
a)	Illustrate the working of RS Flip-Flop with neat diagram and truth table.	10	CO1	BT4
b)	Explain the Internal RAM organization of 8051 and explain it.	10	CO3	BT2
c)	Draw the bit-structure diagram of TMOD and TCON registers and write an assembly language program to generate a delay of 1mS using Timer 0 and Timer Mode-1 of 8051 microcontroller.	10	CO4	BT4